Xiaoshan Wu

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EDUCATION

The University of Hong Kong (HKU) — Hong Kong
Ph.D. in Electrical and Electronic Engineering (Focus: 3D Vision & Generative AI) — *Sep. 2023 – Present*

- **Awards:** Hong Kong PhD Fellowship **(HKPFS, Top Tier)**, HKU Presidential Scholarship.
- **Advisor:** Dr. Xiaojuan Qi. **GPA:** 4.0/4.3.

Zhejiang University (ZJU) & UIUC — China & USA
Dual B.S. in Computer Engineering (ZJUI Institute) — *Sep. 2019 – Jun. 2023*

- **Honors:** Outstanding Graduate (Rank 5), Dean's List. **GPA:** 3.99/4.0 (ZJU), 3.75/4.0 (UIUC).

RESEARCH INTERESTS

My research aims to bridge **3D/4D Vision** to construct next-generation **World Models**. I aim to build systems that are **generative, interactive, persistent, real-time,** and **physically accurate**, leveraging high-fidelity 3D/4D representations to ensure structural consistency in dynamic environment simulation.

SELECTED PUBLICATIONS & PREPRINTS

** indicates equal contribution.*

1. **X. Wu***, Y. Yu*, X. Lyu, et al. **EAG3R: Event-Augmented 3D Geometry Estimation for Dynamic and Extreme-Lighting Scenes**.
   NeurIPS 2025 **(Spotlight, Top 3.18%)**.
2. Y. Yu*, **X. Wu***, X. Hu, et al. **VideoSSM: Autoregressive Long Video Generation with Hybrid State-Space Memory**.
   Under Review at CVPR 2026.
3. **X. Wu**, X. Lyu, Y. Yu, et al. **LiFR-Seg: Anytime High-Frame-Rate Segmentation via Event-Guided Propagation**.
   *Under Review at ICLR 2026 **(6,6,6,4)**.*
4. **X. Wu**, Y. Yu, X. Lyu, et al. **EVSplat: Pose-free Persistent Scene Reconstruction with Event-Visual Motion Disentanglement**.
   Under Review at ICRA 2026.
5. X. Lyu, M. Liu, **X. Wu**, et al. **Stabilizing Streaming Video Geometry via Dynamic Feature Normalization**.
   Under Review at CVPR 2026.
6. M. Liu, X. Lyu, T. Ren, **X. Wu**, et al. **FoundationGeo: Learning Pixel-Wise Scale Fields for Monocular Metric Geometry**.
   Under Review at CVPR 2026.
7. B. Zhang, ... **X. Wu**, et al. **Image2Event: Physics-Aware Event Generation from Static Images**.
   Under Review at CVPR 2026.
8. **X. Wu**, W. He, M. Yao, et al. **Event-based depth prediction with deep spiking neural network**.
   IEEE Transactions on Cognitive and Developmental Systems (TCDS), 2024.
9. H. Chen... **X. Wu**, et al. **Continuous-time digital twin with analog memristive neural ODE solver**.
   Science Advances, 2025.

RESEARCH EXPERIENCE

DiDi AI Research (Autonomous Driving Lab) — Guangzhou, China
Research Intern — *Aug. 2025 – Present*

- Designed an **autoregressive world model** capable of **synchronously predicting RGB, depth, and optical flow**.
- Leveraged these auxiliary geometric and motion cues to enforce physical constraints, significantly improving the temporal consistency of generated videos.
- Mitigated scale ambiguity in monocular settings by aligning visual appearance with geometric predictions.

Megvii Inc. — Beijing, China
Research Intern — *Dec. 2022 – Mar. 2023*

- Developed a **depth-aware semantic segmentation network** for complex autonomous driving scenes.
- Enhanced segmentation accuracy in low-light/occluded scenarios by fusing geometric priors.

TECHNICAL SKILLS

Research Expertise: Long Video Generation, 3D/4D Feedforward Reconstruction, State-Space Models, Linear Attention, Event Vision.
Frameworks & Tools: PyTorch, CUDA, Diffusers, Open3D, OpenCV.

SELECTED AWARDS

Hong Kong PhD Fellowship (HKPFS)	2023
HKU Presidential Scholarship	2023
Outstanding Graduate of Zhejiang University	2023